Date:

Families of FPGAs
- Xilinx
- Actel
- Altera

What FPGA?

- it a PLD with higher densities and capable of implementing
Stands for field Programmable gate Arrays.
- Integrated circuits (ICs)
- Configurable (Programmable)
- it is based on anti-fuse technology

What are programmable components of FPGA?
- logic blocks
- interconnects

- FPGA were introduced in 1985 by Xilinx.
- Very high logic capacity.
Consists of an array of programmable logic blocks surrounded by programmable interconnects.
- Can be configured by end users to implement specific application.
- Capacity up to million logic gates and speed up to 500MHz.

PLA :-

Type of fixed architecture logic device with programmable AND gates followed by programmable OR gate.

PAL :- ~~Pro~~ has programmable OR gate and fixed OR gate
- simple structure
- issue has regarding [illegible]

Liaqat

- A field programmable gate array is a programmable logic device ~~capable~~ with higher densities and capable of implementing different functions in a short period of time.

Whihway to Go?

- off the shelf.
- low development cost
- short time to market
- reprogrammable.

Disadvantage

- Limited performance
- Limited size option
- Not good for high volume applications
- Least efficient use of silicon/wiring resources.

~~Ra~~ Logic Blocks:

Purpose: Implement combinational and sequential logic functions.

Logic block can be implemented by:

- Transistor pairs
- Two input NAND gate or exclusive OR
- Multiplexers
- Look up tables

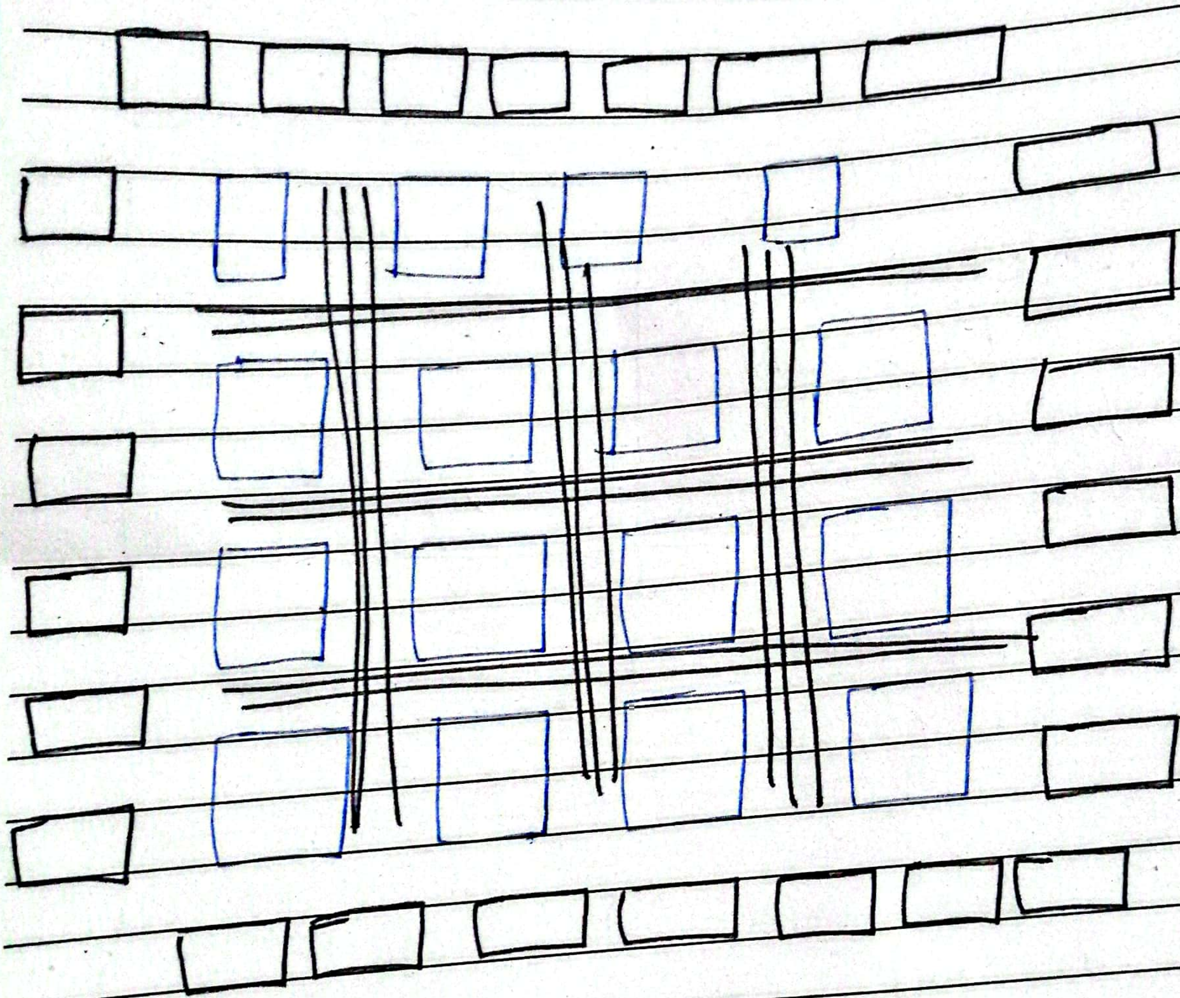
Applications:

- ASCI prototyping
- Software defined equipment

I/O blocks:

- for external connection supports voltages and tri-states.

Block Diagram of FPGA:-



→ FPGA consists of

- Logic elements
- I/O Block
- Programmable interconnect

How does FPGA different from CPLD

CPLD	FPGA
complex programmable logic design.	Field programmable Gate array
it is collection of PLDs.	It is collection of CLBs
PAL like blocks used as PLDs.	CLBs used as building blocks
AND-OR arrays are there in PAL like blocks.	LUTs are there in CLBs.
configuration content is stored in ROM.	configuration content is stored in RAM.
more secure	less secure
cost-effective	Expensive
consume low power	consume more power

what does CLB stands for?
- Configurable logic block.
- implemented in n-input look up table.
- Are arranged in a row and column structure.

- within the CLBs are logic modules joined by local interconnects.
  - Flip-flops
  - LUTS - MUX

Placement of FPGA in digital logic Hierarchy.

- Logic
  - Standard logic
    - Programmable logic devices
      - SPLDs
      - CPLDs
      - $\boxed{\text{FPGAs}}$
      - FICs
  - ASCI
    - Gate Array
    - Cell Band ICs
    - Full custom ICs

- Based on the principle of functional completeness.
- In it, functionally complete elements placed in an interconnect framework.
- Interconnect frame work compraises of wire segments and switches.

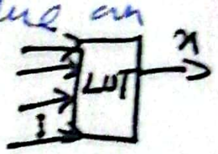
What is difference between local interconnect and a global interconnect in an FPGA?

→ Local interconnect used for short distance on the chip.

→ Global interconnect used for long distances on chip such as clock signal or other controll signal.

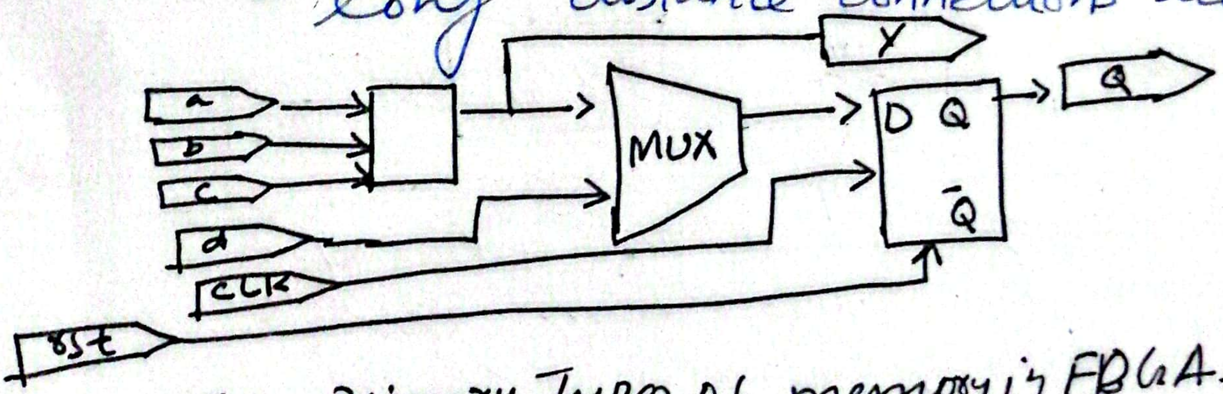
Describe an LUT and its Purpose.

- consist of number of memory cells equal to 2^n where n is the number of i/p variables.
- used to generate SOP combinational logic functions.
- An LUT with three input variable can produce an SOP expression with up to eight product term.



Programmable interconnects:-

- wires to connect input / output and logic blocks
  - Clock
  - Short distance local connections
  - long distance connections across chip



- Two primary Types of memory in FPGA,
  - Distributed memory
  - Block memory

Q# what are logic Modules?

=> A logic module is an FPGA logic block can be configured for combinational logic, registers logic or a combination of both.

=> A flip flop is a part of associated logic and is used for registered logic.

Operating Modes of logic module:

Typically, a logic module can be programmed for the following modes of operation:

- Normal mode
- Extended LUT mode
- Arithmetic mode
- Shared arithmetic mode

Defin intellctual property.

=> Designed owned by a manufacturer of programmable logic devices. A company usually lists the types of intellectual property that available on its website.